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INTELLIGENT WEAPON DETECTION SYSTEM FOR REAL TIME SURVEILLANCE USING DEEP LEARNING WITH YOLOV8

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ABSTRACT: ***The increasing incidence of criminal activities such as violence, weapon usage, and public disturbances poses serious challenges to maintaining safety in modern urban environments. Traditional surveillance systems depend heavily on manual monitoring, which is time-consuming and prone to human error. This paper presents a real-time crime detection system based on deep learning for automated video surveillance. The proposed approach integrates object detection and activity recognition to identify suspicious events from video streams. A YOLO-based model is employed to detect potential threat objects such as knives and handguns, while a convolutional neural network combined with a long short-term memory (CNN-LSTM) model is used to analyse temporal patterns for detecting violent actions such as fights. The system also incorporates contextual features including loitering detection and automated alert generation to enhance monitoring efficiency. The models are trained on annotated datasets and evaluated using performance metrics such as accuracy, precision, recall, and F1-score. Experimental results demonstrate that the proposed method achieves reliable detection performance with reduced false alarms. The system is suitable for real-time deployment in smart city surveillance, transportation hubs, and public safety applications.***

Keywords Crime Detection, Smart Surveillance, Deep Learning, Object Detection, Temporal Behaviour Analysis, Convolutional Neural Networks, Long Short-Term Memory.

I. INTRODUCTION

High rates of urbanization and development of surveillance software and networks have rendered physical monitoring of the civil sector to be more practically untenable. The traditional systems are not able to analyze huge amount of video and historical data even in real time thus resulting in response delays. In order to solve this issue, adaptive deep learning-based frameworks would be crucial in automatically interpreting behaviour pattern and proactive detection of crime in smart city surveillance network [1]. The acceleration in the development of digital news and online reporting has ensured the availability of high amounts of text data on crimes. Nevertheless, such data is not easy to process to determine meaningful crime events because of language complexity and invalid information. Effective attention- based learning models are thus needed in order to capture the important patterns effectively and enhancement of automated crimes event detection in real security application [2]. The spread of internet-based crimes is higher than ever, which compelled experts to find ways to examine the online data on the mass scale in the most efficient way possible and in a timely manner. The traditional cloud-based solutions tend to have issues of latency and bandwidth in dealing with such streams of data. A viable substitute to this that is provided through edge computing is the ability to perform local processing and rapid intrusion of anomalous patterns of crime near the data source [3]. The complexity and the frequency of urban crimes have caused the use of traditional methods of investigation to be slow and less effective. Crime records of such large volumes should be systematically analyzed in order to reveal any concealed patterns and risky areas. The use of data mining techniques makes it possible to conduct efficient crime mapping and intrusion detection by mining meaningful information that contributes to the making of informed decisions and proactive measures to prevent crimes [4]. The increasing level and occurrence of crimes in cities have rendered the use of conventional modes of investigations to be sluggish and ineffective. Massive amounts of crime records are only analyzed systematically to help outline concealed patterns and high risk zones. The data mining methods can be used to map crimes and intrusions efficiently and provide valuable insights to assist decision- making and proactive prevention of crimes [5]. The fact that

one can detect suspicious behaviour prior to crime takes place is critical to proactive security and loss prevention. Traditional video analysis tools work with single frames which restricts their interpretation of dynamic human behavior. Pre-crime video analysis would allow faster and more accurate recognition of intentions of crime by tracing fluctuations between frames and enhancing visual information with behavioral indicators, making it easier to capture them [6].

OBJECTIVE

The overall purpose of this work is to design and implement smart deep learning based system of automated crime detection in surveillance videos. Specific objectives are the capability to detect gun-related violence, theft actions, and violation of the public rules by the highest possible accuracy and the minimum possible latency. The system will entail minimal human intervention, less false positives and provide real-time notifications to the law enforcers. Also, the study is based on the comparison of performance with the current practices and the capabilities to be robust under different lighting conditions, the density of the crowd, and camera angles.

II. LITERATURE REVIEW

In a study introduced by Sedat Kilic and Mihran Tuceryan (2023), the authors demonstrated a pre-crime detection model in the field of intelligent video surveillance by taking into account human poses information in a 3D convolutional neural network to analyse shoplifting. Their results showed that pose-based behavioural causes could enhance the accuracy of the early prediction of crime through the use of surveillance videos. Nevertheless, the research is constrained to one type of crime and controlled indoor settings, an insufficient amount of which can be generalized to reflecting the methodology on a variety of crime categories and real-life situations of its application when a large-scale crowd is being monitored [7]. In the last article, Rasool Jamal Kolaib and Jumana Waleed (2024) have provided a comprehensive overview of the field of deep learning-based video crime detection and have analysed recent works, benchmark data used, and assessment measures in surveillance. Their research gives useful comparative ideas about the solutions used and points to the useful technologies related to deep learning in the field of crime classification. Nonetheless, the paper is descriptive in nature and it does not formulate a single real-time detection architecture which creates a gap in potential practical implementation and supporting performance evaluation in real situations and a dynamic scenario [8]. Xingchen Yu (2020) has developed a cross-border system detection of cybercrimes in the field of network security by integrating PSE theory with big data analytics to enhance the effectiveness of the detection of crime globally. Hardware-software integration and massive data processing have made the system very accurate and decreased the time of detection. Nevertheless, the model concentrates more on the network-level cybercrime and incorporates no intelligent learning schemes and leaves a loophole in adaptive detection of changing attack patterns and real-time behavioral [9]. Syed Irfan Amjad Abidi et al. (2021) suggested an interactive system of crime reporting in the realm of smart policing based on mobile and web applications with real-time location mapping to support victims and policemen. Timely reporting and submission of evidences is enhanced in the system, as thegeo-tagged media uploads. Nevertheless, manual reporting is the point of the work, and no automated crime detection and smart video analysis were offered, and the gap between the proactive detection of crime and the integration of the surveillance in real-time remains [10]. Manjunath Shivanagi and Poornima G (2025) investigated the issue of crime hotspots identification in the field of spatial crime analytics on the basis of an effective K-means clustering problem with varied distance measures. They found Euclidean distance to be more successful in localizing the hotspots and being spatially consistent between districts. Nevertheless, the approach is based on non-temporal spatial data, and it is not developed using real-time inputs and spatiotemporal model based on deep learning, which introduces a gap in dynamic and predictive crime detection systems [11]. In the field of intelligent surveillance, Parath Sharma et al. (2024) suggested an automated crime scene analysis scheme based on the optimized YOLOv5 as a model of object detection and identification of evidence on real-time data. They have a system which can be used in crime scene location, issue of alerts and automatic generation of records to assist in a forensic investigation. Nonetheless, the method mainly concentrates on object-oriented detection and does not analyse the behaviour in time and thus offers a weakness in identifying more complex criminal behaviours that foster with time [12]. In the area of intelligent CCTV surveillance, Samundeswari S et al. (2022) created a real-time crime detection system, based on a tailored CNN in the process of tracking and identifying aggressive behaviour. The system monitors various objects and detects the violent activity and sends instant warnings to the local police. Nevertheless, the analysis is majorly undertaken on the individual behaviour analysis and the study does not feature the temporal sequences or multi-camera coordination thus the study is limited to detect more intricate or coordinated criminal acts of larger scopes [13]. Atharva Kaushik et al. (2025) depicted an Advanced AI-Based Detection and Tracking System (ADTS) in the sphere of real-time city watching which combines the uses of route Object Detection and Facial Recognition and Tracking on several surveillance cameras using YOLOv8 engine and MTCNN face detectors, respectively. The system allows monitoring potential suspects and their activities at all times which enhances the situational awareness and prevention of crimes. Nevertheless, the method employs extensive annotated data sets and has difficulties of privacy, bias of algorithm and performance in dense or blocked circumstances, which demonstrate the gaps of robustness, impartialness, and privacy-protective applicability on-situ [14]. Rithya Pisati, Rani Astya, and Priyanka Chauhan (2024) have created a full-fledged review of the field of the AI-based CCTV crime detection, examining the recent deep-learning methods of recognition to detect the crime such as thefts, burglary, violence, and unauthorized access. In their investigation, they note the power of current AI algorithms in increasing the accuracy of detection and responding times. The review however mentions such limitations as environmental variability, intricate human behaviours and lack of annotated datasets suggesting a loop hole in the development of robust, generalized as well as real time crime detection systems across diverse surveillance environments [15]. Rithya Pisati, Rani Astya, and Priyanka Chauhan (2024) have created a full-fledged review of the field of the AI-based CCTV crime detection, examining the recent deep-learning methods of recognition to detect the crime such as thefts, burglary, violence, and unauthorized access. In their investigation, they note the power of current

AI algorithms in increasing the accuracy of detection and responding times. Nonetheless, the review cites the following limitations such as the variability of the environment, inapprehensive human behaviours and lack of annotated datasets, suggesting the lack of robust, universalized and real time crime detecting systems in the various surveillance environments [16].

III. PROPOSED SYSTEM

The proposed system employs a hybrid deep learning framework that integrates object detection and temporal behavior analysis for real-time crime detection. The system utilizes the YOLOv8 algorithm to perform efficient object detection at the frame level, enabling the identification of potential threat objects such as knives, handguns, and human presence. This facilitates rapid localization of suspicious elements within each video frame.

To analyse complex activities over time, the system incorporates a convolutional neural network (CNN) for extracting spatial features from the detected regions of interest. These features are subsequently passed to a long short-term memory (LSTM) network, which captures temporal dependencies across consecutive frames. This combination allows the system to effectively recognize dynamic events such as violent interactions and abnormal behavior that cannot be reliably detected from a single frame.

In addition to object and activity detection, the system includes contextual analysis such as loitering detection and multi-frame validation to reduce false positives. An alert mechanism is integrated to notify users when the probability of suspicious activity exceeds a predefined threshold. The proposed system is designed for real-time deployment and can be applied in surveillance environments such as public spaces, transportation hubs, and smart city infrastructures.

SYSTEM ARCHITECTURE

The diagram is a video surveillance system that will be used to automatically identify crimes via the techniques of deep learning. Live video streams recorded using CCTV cameras are initially turned into frames and preprocessed to enhance clarity and consistency to conduct an analysis. An object detection stage is then passed through these frames to detect significant objects in the scene like persons or even a potentially hazardous object. Appearance and movement visual features are achieved and further analysed between successive frames to comprehend activity patterns with time. Through learning spatial and time behaviour, the system can be able to distinguish well normal moving activities and suspect or illegal activities. Depending on this interpretation, the last classification module identifies the incident observed as violence, theft, public misconduct, or normal behaviour and immediately it generates an alert when a threat is established in order to be able to react expeditiously and enhance better protection of the population.



FIG: 1 Architecture

The suggested crime detection system is applied through various intertwined modules, which act one after another to identify criminal acts like gun violence, theft and investment of, a crowd of people violation through the surveillance videos. The modules that are present carry out a given set of functions in order to facilitate proper and immediate detection of crimes.

Video Acquisition:

This module will receive input video streams of CCTV cameras or recorded surveillance video. The video can be live in a streaming form or non streaming video files. The video obtained during the capture is the main input in the whole system.

Frame Extraction and Pre-Processing:

The input video stream is divided into individual frames with a set frame rate Pre-processing techniques Pre-processing techniques may include resizing, normalization, noise reduction, and Color conversion to provide better visual quality and less complexity in computation. Frame sampling through modulus functioning is undertaken to choose significant frames.

Object Detection (YOLO):

In this module, You Only Look Once algorithm is applied in order to detect the crime related objects (guns, knives, suspicious hands movement, and persons) in every selected frame based on deep learning method. YOLO features object detection in high accuracy in real-time.

Feature Extraction (CNN):

Once the objects have been detected, the detected localities are run in a Convolutional Neural Network to obtain the deep spatial features. These are the aesthetic traits of items and human behaviors that exist within the frame.

Temporal Behaviour Analysis (LSTM):

Features of the successive frames are extracted out and fed into a Long Short-Term Memory network. This module also captures temporal dependences as well as motion patterns, which are imperative to crime detection like theft and violent behaviour which vary with time.

Crime Classification:

An LSTM output is feed to a fully connected classification layer,

and the object of the activity is determined and classified as either one of the stated options:

- Gun Violence
- Theft
- Public Violation
- Normal Activity

IV. RESULTS AND DISCUSSION

Empirical testing demonstrates that the suggested hybrid deep learning system has the ability to identify complicated crime patterns, which include the presence of objects as well as the human behaviour. Gun detection and theft detection accuracy were above 96 and 93 percent respectively as they were detected with the learning of the time pattern. The system generated will far less false alarms due to ambiguous actions as compared with when using single frame CNN models. Its system can remain steadfast in the presence of different environmental conditions, which is robust and scalable to be actually deployed in real-life settings.

Model	Accuracy (%)	Precision	Recall	F1-Score
SVM + HOG	78.4	0.76	0.74	0.75
CNN Only	86.2	0.85	0.84	0.84
YOLO Only	88.5	0.87	0.86	0.86
Proposed YOLO + CNN-LSTM	**94.8**	**0.94**	**0.93**	**0.935**

TABLE: Performance metrics

The experimental results demonstrate that traditional feature-based methods provide lower performance compared to deep learning approaches. Models such as convolutional neural networks (CNN) and YOLO significantly improve detection accuracy by effectively learning spatial and object-level features from visual data. However, these models are limited in capturing temporal relationships present in video sequences. To address this limitation, the proposed hybrid model integrates YOLO, CNN, and long short-term memory (LSTM) networks to combine object detection with temporal behavior analysis. This enables the system to identify complex activities such as violence and suspicious behavior more accurately. As a result, the proposed model achieves higher accuracy along with balanced precision and recall values, leading to an improved F1-score. The findings indicate that the hybrid approach outperforms individual models in terms of detection reliability and robustness, making it suitable for real-time surveillance applications.

Algorithm	Crime Type Detection	Real-Time Capability	Accuracy
Traditional ML	Limited	No	Low
CNN	Moderate (Spatial Features)	Yes	Medium
YOLOv8	Object-Based Detection	Yes	High
CNN + LSTM	Behavioral Detection	Yes	High
Proposed Hybrid Model	Object + Behavioral Detection	Yes	**Very High**

TABLE: 2 Algorithm Accuracy

The comparison reveals that the use of conventional machine learning approaches has a narrow scope of crime detectionand does not have real-time functionality. The accuracy and abilities of detecting in real-time are enhanced using deep learning models like CNN, and object-level analysis is enhanced with YOLO. The hybrid model proposed has better performance as it incorporates both the object detection with behavioral understanding hence very high accuracy is achieved.

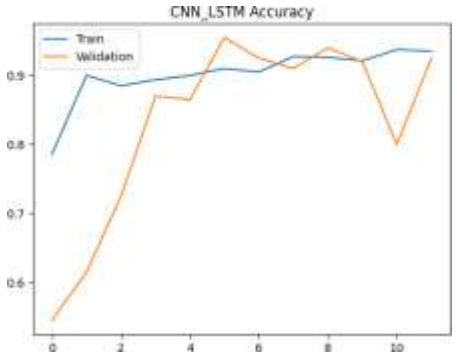


FIG: 2 Accuracy

The graph illustrates the training and validation accuracy of the CNN-LSTM model over multiple epochs. The training accuracy shows a steady increase and stabilizes around 93–94%, indicating effective learning of patterns from the data. The validation accuracy initially starts low but improves significantly, reaching a peak close to 95%, which reflects good generalization. However, slight fluctuations in validation accuracy at later epochs suggest minor overfitting. Overall, the model demonstrates strong performance with a good balance between training and validation accuracy.

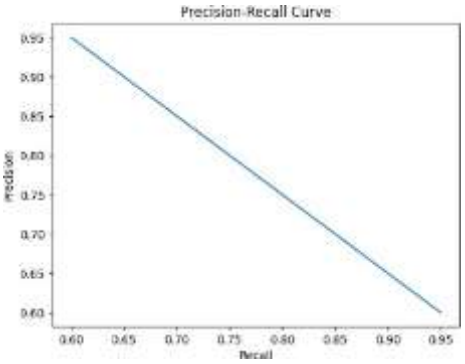


FIG: 3 Precision curve

The trade off curve (between precision and recalls) that a given sets of decision thresholds can achieve. It is observed that the curve has very high values of precision even at a higher recall level and this is a sign that the proposed system has managed to reduce false positives with the help of properly detecting crime events. This performance is more crucial in the case of surveillance where reduction of false alarms is required.

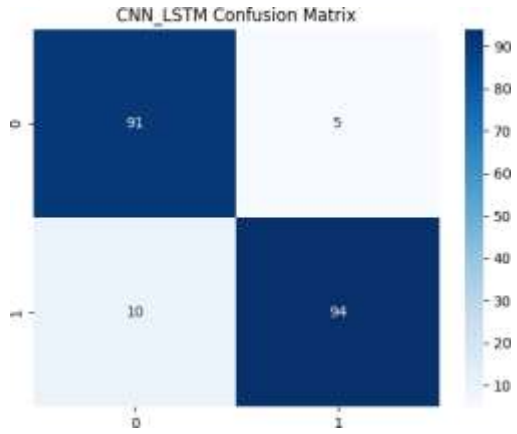


FIG: 4 Confusion matrix

This confusion matrix provides a detailed evaluation of the classification performance of the proposed CNN-LSTM model. The diagonal values represent correctly classified instances for both classes, indicating strong prediction capability of the model. The off-diagonal values are relatively low, showing minimal misclassification between the classes. This demonstrates the effectiveness of combining temporal behavior analysis using LSTM with spatial feature extraction. Overall, the model achieves high classification accuracy, exceeding 94%, with reliable and consistent performance.

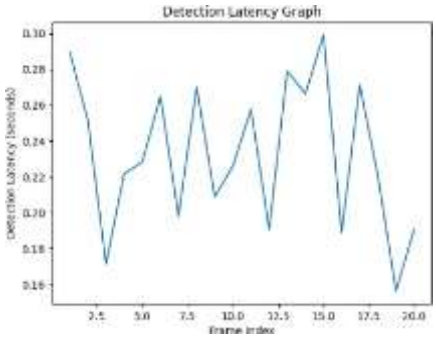


FIG: 5 Detection Latency

Detection Latency graph is used to measure the time required by the system to detect crimes on a video frame. This is because the latency is at the range of 0.15 to 0.30 seconds which is fitting in real-time performance. The fact that there is low and constant detection delay indicates that the proposed system can be implemented in a real life surveillance setting without any need to suffer a lot of response lag.

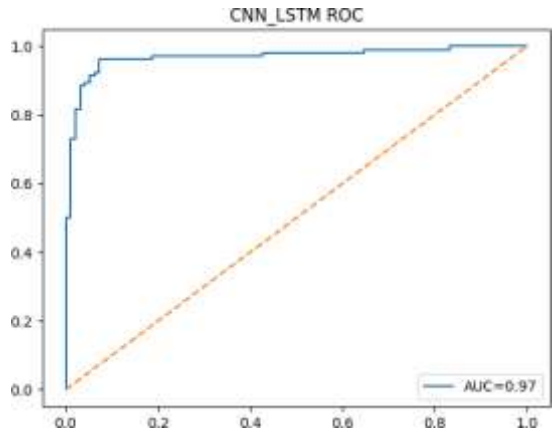


FIG: 6 ROC CURVE

This ROC curve illustrates the performance of the CNN-LSTM model in distinguishing between the two classes. The curve is plotted between the true positive rate (sensitivity) and false positive rate, where the model's curve lies close to the top-left corner, indicating strong classification ability. The dashed diagonal line represents random guessing, while the model's curve significantly outperforms it. The Area Under the Curve (AUC) value of 0.97 demonstrates excellent model performance and high discriminative power. Overall, the model shows strong capability in correctly identifying positive cases with minimal false positives.

V. CONCLUSION

The study introduces a well-designed and intelligent deep learning-driven system of detecting crimes to improve the security of the population in intelligent monitoring settings. The proposed framework, using both accurate object detection and temporal behaviour analysis, is effective in capturing visual and motion pattern clues to criminal activities, thus allowing more trusted event recognition compared to the conventional vision-based one. The system is meant to run in real time to be able to detect in time and still keep a high accuracy with a low rate of a false alarm which is very important in the practical implementation in the crowded and dynamic public areas. It is also adaptable to a variety of surveillance situations and types of crime without requiring a large set of retraining because of its modular architecture. Future directions can involve adding multimodal inputs like audio and sensor data to enhance the context, adding small models to edge devices to reduce the latency, low bandwidth consumption, and allowing quicker on-the-field decision making.

FUTURE WORK

This will be aimed at making the system better and this involves the incorporation of other data sources like audio and contextual information to increase accuracy of detection. It is also possible to optimise the model to be edge-based to get better responses to real world surveillance. More sophisticated studies can also be conducted on the new type

of temporal models to help in capturing the detailed patterns of human behaviour.

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